

# **Proposed Solution**

## **Project Title: Rising Waters**

### **Introduction**

Rising Waters is a Machine Learning-based Flood Prediction System developed to predict the possibility of floods using weather and rainfall parameters. The system provides users with early flood predictions through a simple web application.

### **Proposed Solution**

The proposed solution uses a trained Machine Learning model to analyze important environmental factors such as temperature, humidity, cloud cover, annual rainfall, seasonal rainfall, average June rainfall, and subdivision rainfall. Users enter these values through a web interface, and the system processes the input to predict whether a flood is likely to occur.

The application is developed using Python and Flask, while HTML, CSS, and JavaScript are used to create a user-friendly interface. The trained model generates the prediction and displays the flood probability along with the result.

### **Key Features**

- Predicts flood risk using Machine Learning.
- Accepts weather and rainfall parameters as input.
- Displays Flood or No Flood prediction.
- Shows prediction probability.
- Simple and responsive web interface.

### **Benefits**

- Provides early flood warnings.
- Supports disaster preparedness.
- Reduces manual prediction efforts.
- Easy to use and fast in generating results.

### **Conclusion**

The proposed solution combines Machine Learning and web technologies to provide an efficient flood prediction system. It helps users make informed decisions by providing accurate and timely flood risk predictions.